

# Effective dating — what you asked, what we built

At the second demo (29 September) you asked for the platform to keep history “like an HR system”, to produce HESA returns as a snapshot, and to show programme changes in the return. This note maps each request to what is now built and where to see it in the application.

|                            |                          |                                              |                                        |
|----------------------------|--------------------------|----------------------------------------------|----------------------------------------|
| <b>7</b><br>requests built | <b>1</b><br>partly built | <b>14</b><br>facts now kept as dated history | <b>770+</b><br>automated tests passing |
|----------------------------|--------------------------|----------------------------------------------|----------------------------------------|

## In short

Anything that changes over time is now stored as dated periods instead of being overwritten, with who recorded it and when. Today's value still sits on the student record, so lists and dashboards behave as before. The HESA return reads what was true in each period, can be built as at any date and as the data stood at any moment, and is frozen when it is signed off. Changes that reach a signed-off return are flagged and need the returns team.

## 1. Your requests and what we built

| #          | You asked                                                                      | What we built                                                                                                                                                                                                                                                                                                                                                                  | Where to see it                                                         | Status |
|------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|
| 1.4 / 4.3  | Effective dating across the whole model, “like an HR system”                   | Status, programme, study intensity, module status, fee status, fee eligibility, study location, outside-UK flag, expected end date and unit of assessment are dated facts; funding and supervision are dated periods; custom attributes can opt in. Each entry records who made it and when. A wrong record is corrected (the old entry is kept, superseded), not overwritten. | Student → Dated history tab; Lifecycle changes                          | Built  |
| 1.12       | Return as a snapshot “as at” a date (your 1 August example)                    | Two independent dates: <b>as at</b> (which value was in force on that day) and <b>as known at</b> (only what had been recorded by that moment). Sign-off freezes the exact rows, so later or back-dated changes can't bleed in, and the signed-off version re-downloads unchanged.                                                                                             | Student → As of a date; Statutory → Sign-off → Snapshot as at; Versions | Built  |
| 1.13 / 4.5 | Programme change shown in the statutory return                                 | A student who transfers mid-year appears as one record per programme period, with start and end dates clipped to each programme and the load split between them.                                                                                                                                                                                                               | Statutory → return (CSV / Excel)                                        | Built  |
| —          | Protect returns already submitted                                              | Before approval, a warning names any signed-off return a change would reach. A change dated inside a signed-off year needs a separate permission (returns team only). After sign-off, a list shows every student whose history moved inside that year.                                                                                                                         | Lifecycle request; Statutory → Changes since sign-off                   | Built  |
| 1.3        | Module versioning with effective dates; cohorts keep what they were sold (CMA) | Modules and programmes have versions in force from a date; a version with students on it can't be edited, a change becomes a new version. Each student is pinned to the programme version they started on; module credits and awards use the version actually taken.                                                                                                           | Settings → Programmes (programme and module versions)                   | Built  |

| #   | You asked                                                                   | What we built                                                                                                                                                                                                                          | Where to see it                                                           | Status              |
|-----|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------|
| 1.5 | FTE at module level, and student FTE must not exceed the sum of module FTEs | Module FTE on each version (or derived from credits). Configurable check: off / warn / stop, with a tolerance, research students optional. Runs in the return's validation and when an intensity change is approved (stop refuses it). | Settings → Institution policy → Statutory reporting; Statutory → Validate | <b>Built</b>        |
| UOA | Unit of assessment dated (your supervisor UOA example)                      | A managed UOA list; dated UOA for each student and each supervisor. The return takes both as at the record's date; a supervisor's late change is listed against their students.                                                        | Settings → Units of assessment; person and student pages                  | <b>Built</b>        |
| 1.7 | HESA Engagement entity                                                      | Engagement and Leaver groups in the return, built from the dated history (start, expected end, fee eligibility, research council studentship, leaving reason). Attendance and VLE engagement evidence is not built yet.                | Statutory → field mapping (engagement.*, leaver.*)                        | <b>Partly built</b> |

## 2. Your example, worked through

Alistair's example: a student live in August, interrupted in March, back in June and interrupted again in September. It is set up as a demo student (ED-DEMO-01) in the 2025/26 HESA year, with a drop to 60% study intensity in January as well.

| Date       | What happened                          | Recorded as                             |
|------------|----------------------------------------|-----------------------------------------|
| 1 Aug 2025 | Live, full time                        | Active, 100%                            |
| 5 Jan 2026 | Reduced to 60% alongside clinical work | Intensity change, approved              |
| 2 Mar 2026 | Interrupted                            | Suspension, approved                    |
| 1 Jun 2026 | Back                                   | Return recorded; active again           |
| 7 Sep 2026 | Interrupted again                      | Suspension, approved (falls in 2026/27) |

### What the 2025/26 return says

| Built as at            | Status    | Mode / intensity | Why                                                             |
|------------------------|-----------|------------------|-----------------------------------------------------------------|
| 1 Aug 2025             | Active    | Full time, 100%  | Before any change                                               |
| 1 Apr 2026             | Suspended | Part time, 60%   | Inside the March–June interruption                              |
| 31 Jul 2026 (year end) | Active    | Part time, 60%   | Back since June; the September interruption is outside the year |

#### Load for the year (STULOAD) is day-weighted

100% from August to January, 60% from January to March, nothing while interrupted, 60% from June to July. Over the 365 days of the year that comes to **62.2** (shown as 62 in the return). The status history in the record lists all three periods: active, suspended, active.

A second demo student (ED-DEMO-02) transfers from the ICR PhD to the MD(Res) on 2 February 2026 and appears as two records: PhD from 1 Aug 2025 to 1 Feb 2026 (load 50), MD(Res) from 2 Feb to 31 Jul 2026 (load 49).

### As known at a moment

- Signing off the return stores exactly what it contained at that moment.
- If someone later back-dates a change into the year, the signed-off version is unchanged; the change is flagged as recorded after sign-off and listed under “Changes since sign-off”.
- Making that change needs the returns permission, so it can't happen by accident.

### 3. Still to do

| Item                                                 | Where it stands                                                                                                         |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Amend and resubmit a signed-off return (1.11)</b> | Signed-off versions are frozen and re-downloadable; editing a version with reasons and resubmitting is next.            |
| <b>XML output (1.10)</b>                             | CSV and an Excel review workbook are built; XML against the HESA schema is next.                                        |
| <b>Engagement evidence (1.7)</b>                     | Attendance and VLE activity need a feed from your systems through the integration API; to agree with you.               |
| <b>Fee status codes</b>                              | Fee status is dated; mapping it to HESA FEESTAT codes needs your coding confirmation.                                   |
| <b>FTE check at module enrolment</b>                 | The check runs in the return and on intensity approval; adding modules can't cause a breach, so enrolment only informs. |

#### How it is kept safe

One write path per fact, so nothing can change history behind its back; a database guard against overlapping periods; every institution's history isolated by row-level security; and automated tests for your example, the snapshot rules and every migration.